Secure Multimedia Data Transmission over Cloud Using Multi-Layered Hybrid Encryption

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***Abstract*—With the rapid growth of cloud computing technology, large volumes of multimedia data including images, audio, and video are transmitted and stored in cloud environments. The security and integrity of multimedia data are essential requirements for its secure and efficient transmission. In addition, multimedia data requires a high level of security and integrity to prevent unauthorized access and cyber-attacks during its transmission process. In this regard, the conventional single-layer encryption technique is often inadequate in providing sufficient security against various types of cyber-attacks and threats. In this study, a secure multimedia data transmission system is proposed based on a multi-layered hybrid approach of encryption technology and cloud computing services. In this context, the proposed system has the advantage of efficient multimedia data encryption using AES-256 symmetric encryption algorithm, secure key exchange using RSA-2048 asymmetric encryption algorithm, data integrity verification using SHA-256 hashing algorithm, and digital signatures for authentication purposes. Moreover, the proposed system has the advantage of efficient and secure transmission and storage of the multimedia data in the cloud environment. In order to validate the performance of the proposed multimedia data transmission system, extensive experiments were conducted using the proposed system and multimedia data with different file sizes ranging from kilobytes to more than 100 MB in size. Moreover, the performance metrics of the proposed system were analyzed using performance metrics such as encryption time, cloud upload time, end-to-end transmission time, and system overhead time. The experimental results confirm the effectiveness of the suggested system in terms of efficient multimedia data encryption as well as secure transmission of the data in the cloud environment with minimal computational overhead. This validates the feasibility of the proposed security mechanism in the context of efficient and secure multimedia data transmission in the cloud environment..**

***Index Terms*—Multimedia Security, Hybrid Encryption, AES-256, RSA-2048, Cloud Security, Secure Data Transmission**

I. Introduction

The rapid growth of cloud computing services has increased the rate of multimedia data transmission and storage in networked environments significantly. Multimedia data, such as images, videos, and audio, are commonly used in different applications, including healthcare systems, online education platforms, multimedia streaming services, and social media networks. Although cloud computing services offer data storage and efficient accessibility, it poses different security issues in data transmission and storage. Unauthorized access, data tampering, and interception of sensitive multimedia information are major concerns that must be addressed.

Traditionally, data protection mechanisms have employed single-layer encryption schemes to ensure the security of the data being transferred. Although these methods ensure some level of security, they may not be sufficient to protect multimedia data from the ever-increasing and complex cyber attacks. This is because multimedia data is normally large in size and requires efficient encryption methods that can be achieved without compromising the system's performanceTherefore, designing a secure yet computationally efficient transmission mechanism remains an important research problem in cloud-based systems.

Hybrid methods of encryption have been identified as a way of resolving the trade-offs in communication system security and performance. In hybrid cryptography, symmetric key encryption methods like AES are employed for efficient data encryption, whereas asymmetric key encryption methods like Rivest-Shamir-Adleman (RSA) are employed for efficient key exchange.In addition, hash functions and digital signature schemes can be combined for efficient data integrity and authentication.

In this paper, a secure multimedia transmission system is proposed using a combination of multi-layered cryptography and cloud computing services. In the proposed system, AES-256 is used for multimedia data encryption, RSA-2048 is used for efficient key management, and SHA-256 is used for integrity verification and digital signature schemes for authentication purposes. The proposed system is implemented using the Python programming language and tested using extensive experiments with multimedia data of different sizes.

The performance of the proposed framework is measured based on several parameters, including the time taken for the encryption process, the time taken for uploading the file to the cloud environment, the time taken for the cryptographic operations, the time taken for the transmission of the file from the source to the destination, and the overall system overhead.

Based on the above discussions, the main contributions of the proposed framework can be identified as follows:

• Development of a multi-layered hybrid framework for the encryption of multimedia data.

• Development of the framework based on the AES-256 encryption algorithm, the RSA-2048 encryption algorithm, and the SHA-256 hashing algorithm, and the digital signature scheme.

• Development of a framework for the secure transmission of multimedia data through the cloud environment.

• Evaluation of the system based on the experimental results obtained by using different file sizes.

II. Literature Review

A. Multimedia Security in Cloud

The rapid adoption of cloud computing has significantly increased the volume of multimedia data stored and transmitted through cloud infrastructures. While cloud services provide scalability and accessibility, they also introduce several security concerns including data leakage, unauthorized access, and interception during transmission. Researchers have emphasized the need for strong cryptographic mechanisms to ensure the confidentiality and integrity of multimedia data in cloud environments. For instance, Harnal and Chauhan proposed an end-to-end encrypted multimedia framework to maintain data confidentiality and integrity in cloud systems [1]. Similarly, Saini and Kumar developed a secure multimedia encryption technique designed specifically for protecting multimedia data in cloud environments [2]. Other studies also highlight that cloud-based multimedia systems require secure communication protocols and encryption strategies to mitigate potential cyber threats [3].

B. Hybrid Encryption Systems

The rapid growth of cloud computing technology has increased the rate of multimedia data storage and transmission using cloud computing technology. However, it has also increased various security issues, such as data leakage, unauthorized access, and interception of data during transmission using cloud computing technology. Different researchers have emphasized the need for implementing different types of cryptographic methods to provide security to multimedia data using cloud computing technology. For example, Harnal and Chauhan proposed an end-to-end encryption of multimedia data to provide security to multimedia data using cloud computing technology [1]. In addition, Saini and Kumar have proposed a secure multimedia encryption technique, which is specifically implemented to provide security to multimedia data using cloud computing technology [2]. In addition, different researchers have emphasized the need for designing different types of secure communication and encryption methods to provide security to multimedia data using cloud computing technology against different types of cyber attacks [3].

C. Performance Evaluation Studies

In addition to this, the performance evaluation of encryption techniques has been conducted by the researcher as well. The performance evaluation of encryption techniques is usually conducted based on certain parameters like encryption time, decryption time, computation overhead, and network latency, etc. Murad and Rahouma conducted the performance analysis of hybrid encryption techniques in cloud computing environments and found that the performance of these techniques is highly efficient in minimizing computation overhead [4]. The performance evaluation of the hybrid AES, DES, and RSA encryption algorithms was done by Musa et al. through simulation experiments [5]. Some authors have performed the performance evaluation of the encryption algorithm with respect to the security and performance of the encryption in the cloud computing [6].

D. Research Gap

Despite significant advancements in the development of cryptographic techniques for cloud security, some limitations are observed in existing techniques. Most of the existing works are either focused on encryption algorithm design or cloud storage security, whereas these issues are generally addressed separately in existing literature. Moreover, little work is reported on performance evaluation of combined effects of these security issues on overall performance of systems. Recently, researchers have proposed hybrid encryption systems and multi-layer security systems [7], [8], However, experimental evaluation in terms of encryption time, upload time, encryption processing overhead, and end-to-end transmission latency is limited. Hence, there is a need for a unified framework that encompasses the integration of multi-layered cryptographic security and cloud-based multimedia transmission while analyzing the performance of the system. The proposed work aims at fulfilling this need by implementing the concept of a multi-layered hybrid encryption framework for secure multimedia data transmission over cloud platforms and evaluating its performance through experimental analysis using multimedia data of varied sizes.

III. Proposed System Architecture

To overcome these security issues in the transmission of multimedia data in cloud-based systems, this paper aims to provide a multi-layered hybrid encryption framework to ensure security and privacy of the data transmitted, keeping in mind the performance of the system. The proposed system includes a secure transmission pipeline consisting of a sender module, multi-layered encryption processing, cloud storage, and a receiver module.

The overall architecture of the proposed system is designed to protect multimedia data before it is transmitted to the cloud and to verify its integrity and authenticity when it is retrieved by the receiver. The system employs a combination of symmetric encryption, asymmetric encryption, hashing techniques, and digital signature mechanisms to achieve multiple layers of security.

At the sender side, the multimedia file is first encrypted using the Advanced Encryption Standard (AES-256). AES is a symmetric encryption algorithm known for its high efficiency and strong security, making it suitable for encrypting large multimedia files. The generated AES encryption key is then secured using the RSA-2048 asymmetric encryption algorithm. This ensures that the symmetric key used for encryption can be safely transmitted to the receiver without exposing it to unauthorized parties.

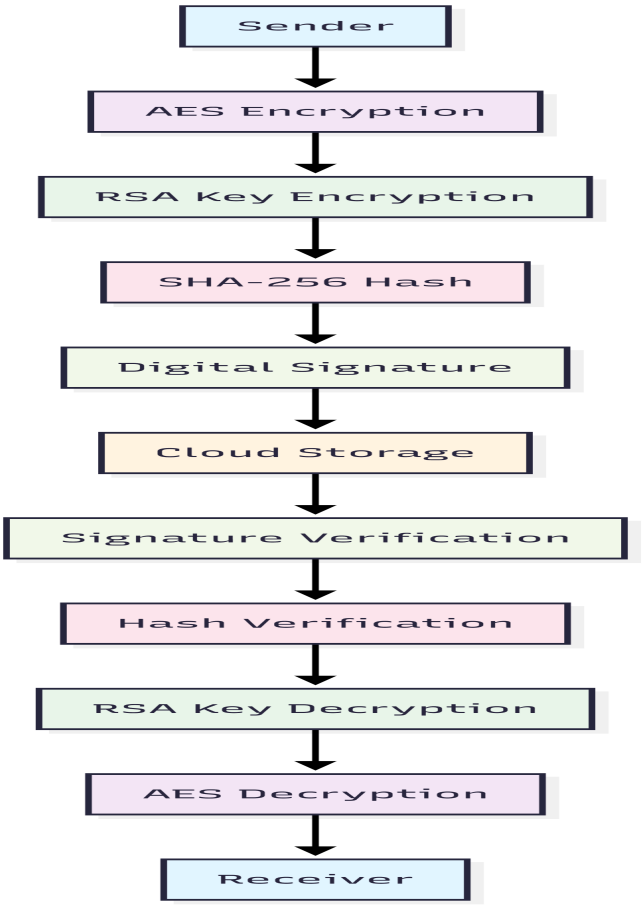

Fig. 1. Proposed Multi-Layer Secure Multimedia Transmission Architecture

To maintain data integrity, a cryptographic hash of the encrypted multimedia file is generated using the SHA-256 hashing algorithm. The hash value acts as a digital fingerprint of the encrypted data and allows the receiver to verify whether the file has been altered during transmission. Additionally, a digital signature is generated using the sender's private key to provide authentication and ensure that the transmitted data originates from a trusted sender.

Once the encryption and security verification processes are completed, the encrypted multimedia file along with its associated metadata (encrypted AES key, hash value, and digital signature) is uploaded to a cloud storage platform. The cloud server acts as an intermediate storage and transmission medium between the sender and the receiver.

At the receiver side, the encrypted multimedia file is retrieved from the cloud storage. The RSA private key is used to decrypt the AES key, which is then used to decrypt the multimedia file. After decryption, the SHA-256 hash of the received file is recomputed and compared with the original hash value to verify data integrity. The digital signature is also verified to confirm the authenticity of the sender.

Through the integration of these cryptographic mechanisms, the proposed architecture provides multiple layers of protection for multimedia data transmitted through cloud environments. The combination of AES encryption for efficient data protection, RSA for secure key exchange, SHA-256 for integrity verification, and digital signatures for authentication ensures that the system achieves strong security while maintaining practical performance for large multimedia files.

IV. Methodology

This section describes the implementation process and experimental setup used to evaluate the proposed multi-layer hybrid encryption framework for secure multimedia data transmission over cloud platforms. The methodology includes the system implementation environment, multimedia dataset used for experimentation, encryption workflow, and performance evaluation metrics.Encrypted multimedia files were uploaded and retrieved through a cloud storage interface to simulate secure transmission between sender and receiver modules.

A. Implementation Environment

The proposed system was implemented using Python programming language due to its extensive cryptographic libraries and ease of integration with cloud services. The cryptographic operations including encryption, decryption, hashing, and digital signatures were implemented using the Python *cryptography* library.

The experiments were conducted on a Windows-based system with the following configuration:

- Processor: Intel Core ultra i5
- RAM: 16 GB
- Operating System: Windows 11
- Programming Language: Python 3.10
- Cryptographic Library: Python cryptography package
- Cloud Storage: API-based cloud storage integration

To improve measurement reliability, each cryptographic experiment was executed three times and the average execution time was recorded as the final performance metric.

B. Multimedia Dataset

To evaluate the performance of the proposed system, multimedia files of varying sizes were used as the experimental dataset. The dataset included image files ranging from small sizes (hundreds of kilobytes) to large multimedia files exceeding 100 MB. Using files of different sizes allowed analysis of how the encryption framework scales with increasing data volume and how it impacts transmission latency and processing time.

C. Encryption Workflow

The secure transmission pipeline follows a multi-layer encryption workflow consisting of the following steps:

1) The sender selects a multimedia file to be transmitted securely.
2) The multimedia file is encrypted using AES-256 symmetric encryption.
3) The generated AES encryption key is encrypted using the RSA-2048 public key.

4) A SHA-256 hash value of the encrypted file is generated to ensure data integrity.
5) A digital signature is created using the sender's private key to authenticate the transmission.
6) The encrypted multimedia file along with the encrypted AES key, hash value, and digital signature is uploaded to cloud storage.
7) The receiver downloads the encrypted file from the cloud.
8) The RSA private key is used to decrypt the AES key.
9) The AES key is then used to decrypt the multimedia file.
10) The SHA-256 hash and digital signature are verified to confirm file integrity and authenticity.

D. Mathematical Representation of Security Model

The proposed secure multimedia transmission framework can be represented mathematically using the following cryptographic operations.

Let the multimedia file be denoted as F and the AES symmetric key be denoted as K_{AES}.

The encrypted multimedia file is generated using AES-256 encryption:

$$C = AES_{K_{AES}}(F) \tag{1}$$

where C represents the encrypted ciphertext.

To securely transmit the AES key, it is encrypted using the receiver's RSA public key:

$$K_{enc} = RSA_{pub}(K_{AES}) \tag{2}$$

To ensure data integrity, a SHA-256 hash of the encrypted file is computed:

$$H = SHA256(C) \tag{3}$$

A digital signature is then generated using the sender's private key:

$$S = Sign_{priv}(H) \tag{4}$$

At the receiver side, the AES key is recovered using RSA decryption:

$$K_{AES} = RSA_{priv}(K_{enc}) \tag{5}$$

Finally, the multimedia file is decrypted using the recovered AES key:

$$F = AES^{-1}_{K_{AES}}(C) \tag{6}$$

Algorithm 1 Secure Multimedia Transmission

1) Input multimedia file F
2) Generate AES symmetric key K_{AES}
3) Encrypt file: $C = AES_{K_{AES}}(F)$
4) Encrypt AES key using RSA public key: $K_{enc} = RSA_{pub}(K_{AES})$
5) Compute hash of encrypted file: $H = SHA256(C)$
6) Generate digital signature: $S = Sign_{priv}(H)$
7) Upload (C, K_{enc}, H, S) to cloud storage
8) Receiver downloads encrypted data
9) Recover AES key: $K_{AES} = RSA_{priv}(K_{enc})$
10) Decrypt multimedia file: $F = AES^{-1}_{K_{AES}}(C)$
11) Verify hash and digital signature

E. Secure Transmission Algorithm

F. Performance Evaluation Metrics

To analyze the efficiency of the proposed system, several performance metrics were measured during the experiments. These include encryption time, decryption time, cryptographic processing time, cloud upload time, and total end-to-end transmission time. Encryption time represents the duration required to encrypt the multimedia file using AES and RSA operations. Cryptographic processing time refers to the total time consumed by all cryptographic operations including hashing and digital signature generation.

Cloud upload time measures the duration required to upload the encrypted multimedia file to the cloud storage platform. End-to-end transmission time represents the total time required for the complete secure transmission pipeline, including encryption, cloud upload, download, and decryption.

All experimental measurements were recorded in structured CSV files to enable systematic analysis and visualization. Cryptographic performance data were stored in $crypto_results.csv, whilecloudtransmissionperformancemetricswere$

V. Experimental Results

This section presents the experimental evaluation of the proposed multi-layer hybrid encryption framework for secure multimedia data transmission over cloud platforms. The performance of the system was analyzed using multimedia files of varying sizes. The experiments measured several performance metrics including cloud upload time, end-to-end transmission time, cryptographic processing time, and system overhead. The recorded measurements were analyzed and visualized using graphical representations.

A. Sample Experimental Measurements

Table 1 presents sample experimental measurements collected during system evaluation. The data includes multimedia file size, cryptographic processing time, cloud upload time, and total end-to-end transmission time. These measurements were extracted from the experimental datasets stored in the CSV files generated during the experimental phase.

TABLE I
SAMPLE EXPERIMENTAL RESULTS

File Size (MB)	Crypto Time (s)	Upload Time (s)	End-to-End Time (s)
0.5	0.12	0.45	0.68
1.0	0.18	0.72	1.03
5.0	0.52	3.10	3.78
20.0	1.40	10.8	12.4
50.0	2.80	25.6	28.7
100.0	4.90	52.3	58.1

B. Comparison with Existing Security Models

Table 2 compares the proposed multi-layer security framework with existing encryption approaches commonly used in cloud-based multimedia systems.

TABLE II
COMPARISON OF SECURITY APPROACHES

Method	Confidentiality	Integrity	Authentication
AES Only	Yes	No	No
AES + RSA Hybrid	Yes	Partial	No
Proposed Multi-Layer Model	Yes	Yes	Yes

C. File Size vs Upload Time

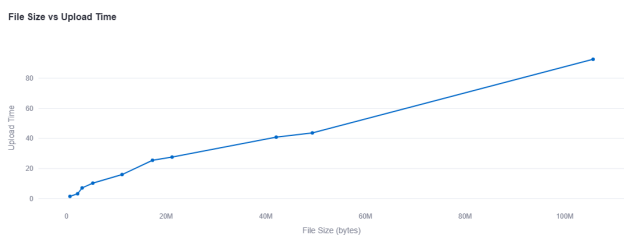

Fig. 2. File Size vs Upload Time

Figure 1 illustrates the relationship between multimedia file size and cloud upload time. The results show that upload time increases proportionally with file size, which is expected due to network bandwidth limitations and cloud storage transfer overhead. Smaller files require minimal upload time, while larger files require longer transmission durations. This observation confirms that the cloud transmission component scales linearly with file size.

D. File Size vs End-to-End Transmission Time

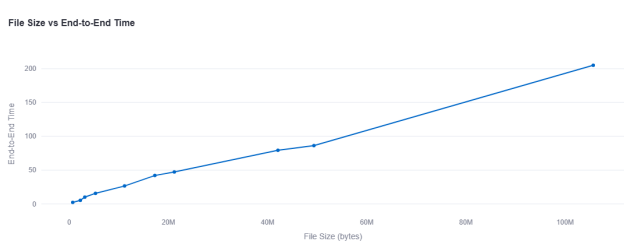

Fig. 3. File Size vs End-to-End Transmission Time

Figure 2 shows the total end-to-end processing time for secure multimedia transmission. This metric includes encryption time, cloud upload time, download time, and decryption time. As file size increases, the total transmission time also increases. However, the results indicate that the major portion of the delay is caused by network transmission rather than cryptographic operations.

E. File Size vs Cryptographic Processing Time

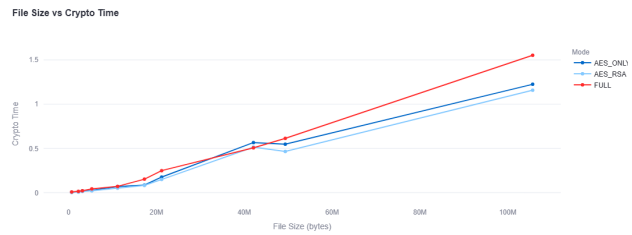

Fig. 4. File Size vs Cryptographic Processing Time

Figure 3 presents the cryptographic processing time required for encryption and related security operations. The results demonstrate that cryptographic processing time increases gradually with file size but remains significantly smaller compared to the overall transmission time. This confirms that the proposed hybrid encryption framework is computationally efficient even for large multimedia files.

F. Cryptographic Contribution to End-to-End Time

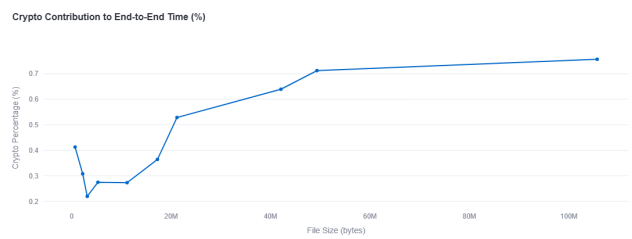

Fig. 5. Cryptographic Contribution to End-to-End Time

Figure 4 illustrates the percentage contribution of cryptographic operations to the total end-to-end transmission time. The results show that cryptographic processing contributes only a very small portion of the overall system latency. This indicates that the proposed multi-layer encryption model introduces minimal computational overhead while providing strong security.

G. File Size vs Encryption Overhead

To evaluate the computational overhead introduced by different security configurations, three experimental setups were considered. The first configuration used only AES encryption for the multimedia file. The second configuration used a hybrid AES-RSA model where AES was used for file encryption and RSA was used for key protection. The third configuration

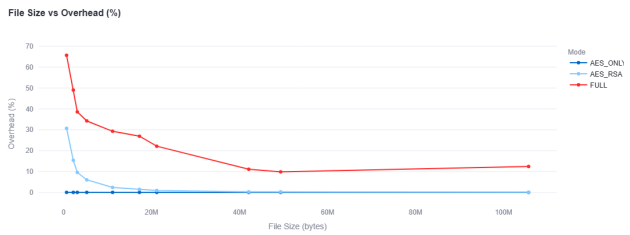

Fig. 6. File Size vs Encryption Overhead

implemented the full multi-layer security pipeline including AES encryption, RSA key protection, SHA-256 hashing, and digital signature generation.

For each configuration, the total cryptographic processing time was measured and compared with the baseline AES-only encryption time. The overhead percentage was calculated to analyze the additional computational cost introduced by the hybrid and multi-layer security models.

The encryption overhead percentage was computed using the following formula:

$$Overhead(\%) = \frac{T_{method} - T_{AES}}{T_{AES}} \times 100 \tag{7}$$

where T_{method} represents the processing time of the evaluated security model and T_{AES} represents the baseline AES encryption time.

Figure 5 compares the encryption overhead introduced by different security configurations including AES-only encryption, hybrid AES-RSA encryption, and the full multi-layer security model. The results demonstrate that although the full system introduces additional security layers, the overhead remains within an acceptable range. This confirms that the proposed framework achieves a balance between security and performance.

VI. Discussion

The experimental evaluation demonstrates that the proposed multi-layer hybrid encryption framework provides strong security while maintaining practical performance for multimedia data transmission over cloud platforms. The results indicate that the overall system performance scales linearly with file size, which is expected when handling large multimedia files in cloud environments.

One of the key observations from the experimental analysis is that cloud transmission time contributes significantly more to the total end-to-end latency than the cryptographic operations themselves. As shown in the experimental results, upload time increases proportionally with file size due to network bandwidth limitations and cloud storage transfer overhead. This suggests that the primary performance bottleneck in cloud-based multimedia systems is network transmission rather than encryption processing.

Another important finding is that the cryptographic processing time remains relatively small even for large multimedia files. The hybrid encryption framework combines AES-256 symmetric encryption for efficient data encryption with RSA-2048 for secure key management. Since AES is optimized for high-speed encryption of large datasets, the additional security layers introduce only minimal computational overhead.

The analysis of cryptographic contribution to total transmission time further supports this observation. The results show that cryptographic operations account for only a small percentage of the overall end-to-end processing time. This confirms that the integration of multiple cryptographic layers does not significantly impact system efficiency.

Additionally, the comparison of encryption overhead across different security configurations shows that while the full multi-layer security model introduces additional processing steps, the overall overhead remains within an acceptable range. This demonstrates that the proposed framework successfully balances security and performance, making it suitable for real-world multimedia transmission scenarios.

Overall, the experimental results validate that the proposed system achieves secure multimedia transmission through a combination of encryption, integrity verification, and authentication mechanisms without introducing significant computational or transmission overhead.

VII. Limitations

Although the proposed framework provides strong security for multimedia transmission over cloud environments, certain limitations remain. The current implementation focuses primarily on file-based multimedia transmission rather than real-time streaming scenarios such as live video or conferencing systems. Additionally, the system performance may vary depending on network bandwidth and cloud storage latency. Future work can address these limitations by exploring optimized encryption pipelines and distributed encryption mechanisms for large-scale cloud environments.

VIII. Conclusion

This paper presented a secure framework for multimedia data transmission over cloud environments using a multi-layer hybrid encryption approach. The proposed system integrates AES-256 symmetric encryption, RSA-2048 based key protection, SHA-256 hashing for data integrity, and digital signature mechanisms for authentication. By combining these cryptographic techniques, the framework ensures confidentiality, integrity, and authenticity of multimedia data during cloud transmission.

The system was implemented using Python and evaluated using multimedia files of varying sizes to analyze the performance impact of the proposed security architecture. Experimental results demonstrate that while the total transmission time increases with file size, the majority of the delay is attributed to network transmission rather than cryptographic operations. The results further show that cryptographic processing contributes only a small fraction of the total end-to-end processing time, indicating that the proposed security model introduces minimal computational overhead.

The analysis of encryption overhead and cryptographic processing time confirms that the hybrid encryption framework achieves an effective balance between security and system efficiency. Even with the integration of multiple security layers, the framework remains practical for real-world multimedia transmission scenarios. Overall, the proposed approach provides a robust and scalable solution for securing multimedia data in cloud-based environments.

IX. Future Work

Although the proposed multi-layer hybrid encryption framework provides strong security and efficient performance for multimedia data transmission over cloud environments, several improvements can be explored in future research. One possible extension is the integration of elliptic curve cryptography (ECC) as an alternative to RSA for key exchange, which may further reduce computational overhead while maintaining strong security guarantees.

Another potential improvement is the implementation of distributed or edge-based encryption mechanisms to reduce network latency and improve system scalability. In addition, future work may investigate the integration of blockchain-based storage or verification mechanisms to provide enhanced data integrity and traceability in cloud environments.

The current system primarily focuses on file-based multimedia transmission. Future studies could extend the framework to support real-time multimedia streaming applications such as video conferencing and live media broadcasting, where low latency and high security are both critical requirements.

Furthermore, evaluating the proposed framework on different cloud platforms and large-scale distributed environments would provide deeper insights into its scalability and real-world deployment potential. These extensions could further enhance the applicability of the proposed security framework for next-generation cloud-based multimedia systems.

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